

Multimodal YFB: Derivation and Updates

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1 Executive summary

Throughout, $i = 1, \dots, n$ indexes matched subjects, $m \in \{1, 2\}$ indexes modalities, $j = 1, \dots, p_m$ indexes features within modality m , and $k = 1, \dots, K$ indexes factors. The shorthand $\sum_{m,j}$ means $\sum_{m=1}^2 \sum_{j=1}^{p_m}$. When used in factor sums, a also ranges from 1 to K ; superscripts $-k$ and $-(m, j)$ indicate that factor k or feature (m, j) has been omitted.

This proposal defines a joint supervised EBMF–Cox model for matched expression and methylation. The modalities share subject scores L and survival coefficients β , while their loading matrices F_1, F_2 , point-exponential empirical-Bayes priors, and feature-wise residual precisions are modality-specific. The linear predictor $\eta_i = \sum_k \beta_k \sum_{m,j} y_{mij} f_{mjk}$ makes the Cox partial likelihood directly inform the loading updates. Thus, the model is not an unsupervised EBMF fit followed by a Cox regression: survival can influence which modality-specific loading patterns are retained.

With the Cox working quantities fixed, the main updates are:

- **Shared scores L .** Both reconstruction blocks contribute $A_{L_{ik}}$ and $B_{L_{ik}}$; their point-exponential EBNM update pools subjects within a factor. YFB has no direct survival term in this update.
- **Loadings F_m .** Each loading has reconstruction and survival contributions to $A_{F_{mjk}}$ and $B_{F_{mjk}}$, followed by a modality–factor-specific point-exponential EBNM update. Coordinates are updated sequentially because the survival predictor couples features.
- **Survival coefficients β .** Cox contributions define A_{β_k} and B_{β_k} ; the Normal prior gives a direct conjugate posterior update.
- **Residual precisions τ_{mj} .** Each feature receives the feature-wise maximum-likelihood update based on its expected squared reconstruction residual.

For all EBNM updates, the quadratic terms give the pseudo-observation $x = B/A$ with standard-error squared $s^2 = A^{-1}$. An intentionally over-specified K_{init} is reduced to an effective factor count K_{eff} through factor-wise empirical-Bayes shrinkage. The following sections state the full objective, the exact A/B expressions, and the update order.

2 Model and priors

For matched subjects, let $Y_m \in \mathbb{R}^{n \times p_m}$ denote expression ($m = 1$) or methylation ($m = 2$). The extension stacks features while retaining one set of subject scores. Write $Y_m = (y_{mij})$ and $E_m = (E_{mij})$:

$$Y = [Y_1 \ Y_2], \quad F = \begin{bmatrix} F_1 \\ F_2 \end{bmatrix}, \quad Y_m = LF_m^\top + E_m. \quad (1)$$

- $L \in \mathbb{R}^{n \times K}$: shared subject scores.
- $F_m \in \mathbb{R}^{p_m \times K}$: modality-specific loadings.
- $E_{mij} \sim N(0, \tau_{mj}^{-1})$: independent Gaussian errors; each feature has its own precision.
- $\beta \in \mathbb{R}^K$: shared survival coefficients (the code's **B/EBeta**).

The **linear joint-YFB** survival predictor is

$$Z_{ik} = \sum_{m=1}^2 \sum_{j=1}^{p_m} y_{mij} f_{mjk}, \quad \boxed{\eta_i = \sum_k Z_{ik} \beta_k.} \quad (2)$$

Unlike LB, whose predictor is $L\beta$, YFB depends directly on F and β . Thus survival contributes to their updates, but not directly to L .

Notation correspondence to the current single-modality YFB model. The posterior moments denoted in code by **EL**, **EL2**, **EF**, **EF2**, **EBeta**, **EBeta2**, and **Tau** correspond here to $\bar{L}$, $L^{(2)}$, $\bar{F}$, $F^{(2)}$, $\bar{\beta}$, $\beta^{(2)}$, and τ , respectively. The single-modality projection **ZF = Y %*% EF** is $\bar{Z}_{ik} = \sum_j y_{ij} \bar{f}_{jk}$; its multimodal analogue is $\bar{Z}_{ik} = \sum_{m=1}^2 \sum_j y_{mij} \bar{f}_{mjk}$. The code's partial residual **R_k** is the same object written here as R^{-k} , and its Cox working response $z_i = \eta_i^0 + u_i/w_i$ equals h_i/w_i when $w_i > 0$. Earlier single-mode derivations write the coefficient as $\tilde{\beta}$; it is the same quantity written here as β . The raw multimodal projection and its nonzero survival contribution to F are intentional proposed-model differences, not changes in the meanings of these shared symbols.

The empirical-Bayes priors are independent across entries, conditional on column-specific distributions:

$$\begin{aligned} l_{ik} &\sim g_{L_k} = (1 - \pi_{L_k})\delta_0 + \pi_{L_k} \text{Exp}(r_{L_k}), \\ f_{mjk} &\sim g_{F_{mk}} = (1 - \pi_{F_{mk}})\delta_0 + \pi_{F_{mk}} \text{Exp}(r_{F_{mk}}), \\ \beta_k &\sim g_{\beta_k} = N(0, s_{\beta_k}^2). \end{aligned} \quad (3)$$

Here π is the nonzero probability and r is the exponential rate. Every modality-factor pair has its own $(\pi_{F_{mk}}, r_{F_{mk}})$, fitted across that modality's features. The K score columns are shared, but a loading column may vanish in one modality. Hyperparameters are estimated; no hyperpriors or precision priors are introduced.

This proposal uses nonnegative point-exponential factors on a prespecified analysis scale. Each block must therefore be prepared on a documented nonnegative scale: for example, normalized nonnegative expression abundance and beta-scale methylation. Simulations must use the same convention. A transformation that recenters features below zero is outside this formulation; using one would require signed factor priors and a new derivation. The raw linear projection in Equation 2 differs from the normalized single-modality YFB implementation and permits the joint loading update below.

The proposed default is the literal unweighted joint likelihood: each observation contributes once, with modality-specific precisions and priors. This does not force expression and methylation to have

equal aggregate influence when $p_2 \gg p_1$. Feature-count imbalance will therefore be evaluated in simulation; inverse-feature-count weighting is a separate model extension, not an implicit default.

3 EBMF plus survival objective

Let t_i be observed survival time and δ_i the event indicator. At distinct event time t_r , write D_r for the event set, $d_r = |D_r|$, and R_r for the risk set; here r indexes distinct event times, whereas r_{L_k} and $r_{F_{mk}}$ above are exponential rates. The Breslow Cox partial likelihood is

$$\ell_C(\eta) = \sum_r \left\{ \sum_{i \in D_r} \eta_i - d_r \log \left(\sum_{i \in R_r} e^{\eta_i} \right) \right\}. \quad (4)$$

The EBMF observation likelihood becomes a sum over modalities:

$$\ell_Y = \sum_{m=1}^2 \left\{ \frac{n}{2} \sum_j \log \tau_{mj} - \frac{1}{2} \sum_{ij} \tau_{mj} \left(y_{mij} - \sum_k l_{ik} f_{mjk} \right)^2 - \frac{np_m}{2} \log(2\pi) \right\}. \quad (5)$$

The joint estimation criterion multiplies the observation likelihood, Cox partial likelihood, and independent priors:

$$\exp(\ell_Y + \ell_C) \left[\prod_{ik} g_{L_k}(l_{ik}) \right] \left[\prod_{mjk} g_{F_{mk}}(f_{mjk}) \right] \left[\prod_k g_{\beta_k}(\beta_k) \right].$$

The survival factor is a **partial likelihood**, not a full likelihood for survival times and a baseline hazard.

Use the mean-field approximation $q = \prod_{ik} q(l_{ik}) \prod_{mjk} q(f_{mjk}) \prod_k q(\beta_k)$. For fixed priors and precision, its objective is

$$\begin{aligned} \mathcal{J}(q, g, \tau) = & \mathbb{E}_q[\ell_Y] + \mathbb{E}_q[\ell_C] - \sum_{ik} \text{KL}(q(l_{ik}) \| g_{L_k}) \\ & - \sum_{mjk} \text{KL}(q(f_{mjk}) \| g_{F_{mk}}) - \sum_k \text{KL}(q(\beta_k) \| g_{\beta_k}). \end{aligned} \quad (6)$$

Relative to single-modality EBMF plus survival, the reconstruction term and loading-prior penalty now sum over modalities. There is still only one L prior term and one β prior term. All likelihood multipliers are one, so the F update below **adds** its reconstruction and survival terms rather than convex-combining them.

The current single-modality YFB implementation instead sets `alpha_F = 0`, suppressing the survival contribution to F because earlier dual-source updates were unstable. That stability safeguard is not part of the proposed multimodal model. Restoring stable survival feedback to the current single-modality YFB model is a separate model-development task.

Write $\bar{\theta} = \mathbb{E}_q \theta$, $\theta^{(2)} = \mathbb{E}_q \theta^2$, and $v_\theta = \theta^{(2)} - \bar{\theta}^2$. Each scalar update below isolates a quadratic $-A_\theta \theta^2/2 + B_\theta \theta$ and hence has the form

$$\boxed{x_\theta = B_\theta/A_\theta, \quad s_\theta^2 = A_\theta^{-1}, \quad q(\theta) \propto g(\theta)N(x_\theta; \theta, s_\theta^2).} \quad (7)$$

These are empirical-Bayes normal-means (EBNM) inputs. Second moments must be retained along with means. Updates with $A_\theta = 0$ are treated directly from the prior when $B_\theta = 0$, rather than dividing by zero. Here and below, C denotes terms constant with respect to the coordinate being updated.

4 Cox working approximation

To obtain quadratic updates, expand at the current mean predictor η^0 . Define risk-set probabilities, score, and working curvature by

$$P_{ri} = \frac{\mathbf{1}\{i \in R_r\}e^{\eta_i^0}}{\sum_{a \in R_r} e^{\eta_a^0}}, \quad u_i = \delta_i - \sum_r d_r P_{ri}, \quad w_i = \sum_r d_r P_{ri}. \quad (8)$$

The exact negative Hessian is $\sum_r d_r \{\text{diag}(P_r) - P_r P_r^\top\}$. We retain the existing **diagonal working approximation** $W = \text{diag}(w)$; w_i is not the exact Hessian diagonal. Replacing the curvature gives

$$\begin{aligned} Q_C(\eta \mid \eta^0) &= \ell_C(\eta^0) + u^\top(\eta - \eta^0) - \frac{1}{2}(\eta - \eta^0)^\top W(\eta - \eta^0) \\ &= C + \sum_i h_i \eta_i - \frac{1}{2} \sum_i w_i \eta_i^2, \quad h_i = u_i + w_i \eta_i^0. \end{aligned} \quad (9)$$

For $w_i > 0$, let $z_i^* = h_i/w_i$; then $Q_C = C' - \frac{1}{2} \sum_i w_i (z_i^* - \eta_i)^2$. The division-free form using h_i also applies at zero curvature. Hold (w, h) fixed during a coordinate sweep.

To take expectations, independence of loading entries gives

$$\bar{Z}_{ik} = \sum_{mj} y_{mij} \bar{f}_{mjk}, \quad v_{Z_{ik}} = \sum_{mj} y_{mij}^2 v_{f_{mjk}}, \quad Z_{ik}^{(2)} = \bar{Z}_{ik}^2 + v_{Z_{ik}}. \quad (10)$$

Independence of factors and of F from β then gives

$$\mu_{\eta_i} = \sum_k \bar{\beta}_k \bar{Z}_{ik}, \quad v_{\eta_i} = \sum_k \left\{ \beta_k^{(2)} v_{Z_{ik}} + v_{\beta_k} \bar{Z}_{ik}^2 \right\}. \quad (11)$$

Consequently,

$$\mathbb{E}_q Q_C = C + \sum_i h_i \mu_{\eta_i} - \frac{1}{2} \sum_i w_i (\mu_{\eta_i}^2 + v_{\eta_i}). \quad (12)$$

In the following updates, replace $\mathbb{E} \ell_C$ in Equation 6 by this expectation. The updates optimize the resulting **fixed working objective**; refreshing the expansion does not guarantee monotone improvement in the original nonquadratic variational objective.

5 Update the shared scores L

Isolate the terms. For coordinate l_{ik} , define the residual excluding factor k ,

$$R_{mij}^{-k} = y_{mij} - \sum_{a \neq k} l_{ia} f_{mja}, \quad \bar{R}_{mij}^{-k} = y_{mij} - \sum_{a \neq k} \bar{l}_{ia} \bar{f}_{mja}. \quad (13)$$

Only reconstruction and g_{L_k} involve this coordinate:

$$\log q^*(l_{ik}) = -\frac{1}{2} \sum_{mj} \tau_{mj} \mathbb{E}_{-l_{ik}} [(R_{mij}^{-k} - l_{ik} f_{mjk})^2] + \log g_{L_k}(l_{ik}) + C. \quad (14)$$

Here $q^*(l_{ik})$ is the coordinate-optimal variational factor and $\mathbb{E}_{-l_{ik}}$ takes expectation over every variational variable except l_{ik} .

Expand and take expectations. Terms depending on l_{ik} are

$$-\frac{1}{2} \sum_{mj} \tau_{mj} \left\{ l_{ik}^2 f_{mjk}^{(2)} - 2l_{ik} \bar{f}_{mjk} \bar{R}_{mij}^{-k} \right\}.$$

The cross-product factors because R^{-k} excludes factor k . Both modalities therefore contribute information about the same score.

Identify coefficients and update.

$$\begin{aligned} A_{L_{ik}} &= \sum_{m=1}^2 \sum_j \tau_{mj} f_{mjk}^{(2)}, \\ B_{L_{ik}} &= \sum_{m=1}^2 \sum_j \tau_{mj} \bar{f}_{mjk} \bar{R}_{mij}^{-k}, \\ x_{L_{ik}} &= B_{L_{ik}} / A_{L_{ik}}, \quad s_{L_{ik}}^2 = A_{L_{ik}}^{-1}. \end{aligned}$$

(15)

For a fixed column k , the subject updates separate. Estimate its prior using the EBNM marginal likelihood

$$\hat{g}_{L_k} = \arg \max_{g \in \mathcal{G}_{\text{PE}}} \sum_i \log \int N(x_{L_{ik}}; l, s_{L_{ik}}^2) dg(l), \quad (16)$$

then obtain $\bar{l}_{ik}$ and $l_{ik}^{(2)}$ from Equation 7 with $\hat{g}_{L_k}$. Here $\mathcal{G}_{\text{PE}}$ is the point-exponential family in Equation 3.

What changes: the sums include all modality blocks. The EBNM step and absence of a direct survival term in the YFB score update are unchanged. There is no separate score matrix for methylation.

6 Update modality-specific loadings F_m

Reconstruction terms. Isolating $f = f_{mjk}$ yields $-\frac{1}{2}\tau_{mj}\sum_i \mathbb{E}(R_{mij}^{-k} - l_{ik}f)^2$. Expanding gives $-A_Y f^2/2 + B_Y f$, where

$$A_Y = \tau_{mj} \sum_i l_{ik}^{(2)}, \quad B_Y = \tau_{mj} \sum_i \bar{l}_{ik} \bar{R}_{mij}^{-k}. \quad (17)$$

Survival terms. Separate other factors from other features of this factor:

$$\begin{aligned} \eta_i &= \eta_i^{-k} + \beta_k(y_{mij}f + Z_{ik}^{-(m,j)}), & \eta_i^{-k} &= \sum_{a \neq k} \beta_a Z_{ia}, \\ Z_{ik}^{-(m,j)} &= Z_{ik} - y_{mij}f_{mjk}, & \bar{Z}_{ik}^{-(m,j)} &= \bar{Z}_{ik} - y_{mij}\bar{f}_{mjk}. \end{aligned} \quad (18)$$

Let $h_i^{-k} = h_i - w_i \bar{\eta}_i^{-k}$, where $\bar{\eta}_i^{-k} = \sum_{a \neq k} \bar{\beta}_a \bar{Z}_{ia}$. Thus, $\bar{Z}_{ik}^{-(m,j)}$ is the mean projection for factor k with feature j of modality m omitted. In Equation 9, the terms depending on f are

$$\sum_i h_i^{-k} \bar{\beta}_k y_{mij} f - \frac{1}{2} \sum_i w_i \beta_k^{(2)} \{y_{mij}^2 f^2 + 2y_{mij} f \bar{Z}_{ik}^{-(m,j)}\}. \quad (19)$$

The second cross-product uses $\mathbb{E}\beta_k^2$, since both features multiply the **same** random coefficient. It includes features from both modalities.

Combine coefficients and apply the modality's prior.

$$\begin{aligned} A_{F_{mjk}} &= \tau_{mj} \sum_i l_{ik}^{(2)} + \beta_k^{(2)} \sum_i w_i y_{mij}^2, \\ B_{F_{mjk}} &= \tau_{mj} \sum_i \bar{l}_{ik} \bar{R}_{mij}^{-k} \\ &\quad + \sum_i y_{mij} \left\{ \bar{\beta}_k h_i^{-k} - w_i \beta_k^{(2)} \bar{Z}_{ik}^{-(m,j)} \right\}, \\ x_{F_{mjk}} &= B_{F_{mjk}} / A_{F_{mjk}}, \quad s_{F_{mjk}}^2 = A_{F_{mjk}}^{-1}. \end{aligned} \quad (20)$$

Use Equation 7 with $g_{F_{mk}}$ to obtain the two posterior moments. Survival couples loading coordinates through $\bar{Z}^{-(m,j)}$. Therefore update coordinates sequentially using current moments, holding the prior fixed during the sweep; independent simultaneous featurewise EBNM fits would not be this coordinate algorithm.

Thus, unlike an unsupervised EBMF fit followed by Cox regression, the Cox partial likelihood directly informs the loading patterns selected in both modality blocks.

6.1 Estimate each loading prior

After updating the loadings of modality m and factor k , hold their posteriors fixed and maximize the expected log prior. Let $\omega_{mjk} = q(f_{mjk} > 0)$ be the posterior slab probability. For $\pi = \pi_{F_{mk}}$ and $r = r_{F_{mk}}$, the relevant terms are

$$\sum_j \{\omega_{mjk} \log \pi + (1 - \omega_{mjk}) \log(1 - \pi)\} + \left(\sum_j \omega_{mjk} \right) \log r - r \sum_j \bar{f}_{mjk}. \quad (21)$$

Differentiating with respect to π and r gives

$$\frac{\sum_j \omega_{mjk}}{\pi} - \frac{p_m - \sum_j \omega_{mjk}}{1 - \pi} = 0, \quad \frac{\sum_j \omega_{mjk}}{r} - \sum_j \bar{f}_{mjk} = 0.$$

Thus the empirical-Bayes coordinate maximization is

$$\boxed{\pi_{F_{mk}} \leftarrow \frac{1}{p_m} \sum_j \omega_{mjk}, \quad r_{F_{mk}} \leftarrow \frac{\sum_j \omega_{mjk}}{\sum_j \bar{f}_{mjk}}.} \quad (22)$$

These equations are applied **separately to expression and methylation**. If the posterior slab mass is zero, the fitted distribution is δ_0 ; the unused slab rate is unidentified. The next loading sweep uses the updated prior. A posterior/prior sweep may be repeated to convergence at fixed working curvature.

Why this prior step differs from the score step. Given other factors, the L likelihood separates over subjects, allowing a vector EBNM fit. The survival likelihood for F couples features. Alternating scalar posterior updates and the prior maximization above respects that coupling. Each scalar posterior uses a common modality-factor prior, not a separately estimated prior for every loading.

What changes from a single modality: each loading gets information from its own reconstruction block and the shared survival projection; prior estimation pools only features in its own modality. Setting the survival contribution to zero would produce an omics-only loading update, which is not the joint model derived here.

7 Update the shared survival coefficients β

Isolate the terms. Write $\eta_i = \eta_i^{-k} + Z_{ik}\beta_k$. The working survival term and g_{β_k} are the only terms involving β_k :

$$\log q^*(\beta_k) = \sum_i \left\{ h_i^{-k} \mathbb{E} Z_{ik} \beta_k - \frac{1}{2} w_i \mathbb{E} Z_{ik}^2 \beta_k^2 \right\} + \log g_{\beta_k}(\beta_k) + C. \quad (23)$$

Here Z_{ik} is independent of η_i^{-k} under the mean-field approximation. The second moment follows from Equation 10, including loading uncertainty in both modalities.

Identify coefficients.

$$\begin{aligned}
A_{\beta_k} &= \sum_i w_i \left(\bar{Z}_{ik}^2 + v_{Z_{ik}} \right), \\
B_{\beta_k} &= \sum_i \bar{Z}_{ik} h_i^{-k}, \\
x_{\beta_k} &= B_{\beta_k} / A_{\beta_k}, \quad s_{\text{obs},k}^2 = A_{\beta_k}^{-1}.
\end{aligned} \tag{24}$$

Estimate the prior and obtain posterior moments. With prior variance $s_{\beta_k}^2$, the scalar marginal distribution is $x_{\beta_k} \sim N(0, s_{\text{obs},k}^2 + s_{\beta_k}^2)$. Differentiating its log likelihood with respect to the total variance gives $-1/(2v) + x_{\beta_k}^2/(2v^2) = 0$. Enforcing a nonnegative prior variance gives

$$\hat{s}_{\beta_k}^2 = \max\{0, x_{\beta_k}^2 - s_{\text{obs},k}^2\}. \tag{25}$$

For $\hat{s}_{\beta_k}^2 > 0$, completing the square in $-A_{\beta_k} \beta_k^2/2 + B_{\beta_k} \beta_k - \beta_k^2/(2\hat{s}_{\beta_k}^2)$ gives

$$v_{\beta_k} = \left(A_{\beta_k} + \hat{s}_{\beta_k}^{-2} \right)^{-1}, \quad \bar{\beta}_k = v_{\beta_k} B_{\beta_k}, \quad \beta_k^{(2)} = v_{\beta_k} + \bar{\beta}_k^2. \tag{26}$$

At zero fitted prior variance, $q(\beta_k) = \delta_0$. The coefficient remains shared; there is no expression-specific or methylation-specific β_k . Replacing $Z^{(2)}$ by $\bar{Z}^2$ would omit loading uncertainty and would not yield this variational update.

8 Update feature-wise precisions τ_m

Isolate the terms. For a single feature, define $S_{mij} = \mathbb{E}_q[(y_{mij} - \sum_k l_{ik} f_{mjk})^2]$. The precision-dependent objective is

$$\mathcal{J}(\tau_{mj}) = \frac{n}{2} \log \tau_{mj} - \frac{\tau_{mj}}{2} \sum_i S_{mij} + C. \tag{27}$$

Expand and take expectations. For $X_{mij} = \sum_k l_{ik} f_{mjk}$, $\mathbb{E}(y - X)^2 = (y - \mathbb{E}X)^2 + \text{Var } X$. Independence between L and F , and across factor columns, implies

$$\begin{aligned}
\mathbb{E}X_{mij} &= \sum_k \bar{l}_{ik} \bar{f}_{mjk}, \\
\text{Var } X_{mij} &= \sum_k \left\{ l_{ik}^{(2)} f_{mjk}^{(2)} - \bar{l}_{ik}^2 \bar{f}_{mjk}^2 \right\}.
\end{aligned}$$

Therefore the expected squared residual is

$$S_{mij} = \left(y_{mij} - \sum_k \bar{l}_{ik} \bar{f}_{mjk} \right)^2 + \sum_k \left\{ l_{ik}^{(2)} f_{mjk}^{(2)} - \bar{l}_{ik}^2 \bar{f}_{mjk}^2 \right\}. \tag{28}$$

The variance term can equivalently be written $\sum_k \{v_{l_{ik}} \bar{f}_{mjk}^2 + v_{f_{mjk}} \bar{l}_{ik}^2 + v_{l_{ik}} v_{f_{mjk}}\}$. This form makes its nonnegativity explicit.

Differentiate and solve.

$$\frac{\partial \mathcal{J}}{\partial \tau_{mj}} = \frac{n}{2\tau_{mj}} - \frac{1}{2} \sum_i S_{mij} = 0, \quad \frac{\partial^2 \mathcal{J}}{\partial \tau_{mj}^2} = -\frac{n}{2\tau_{mj}^2} < 0.$$

For positive expected residual sum of squares,

$$\boxed{\tau_{mj} \leftarrow \frac{n}{\sum_i S_{mij}}, \quad m = 1, 2, \quad j = 1, \dots, p_m.} \quad (29)$$

This is a point maximum-likelihood update, not a Gamma posterior update. It has the same form as single-modality EBMF, applied independently to features in each block. If the expected residual sum is zero, no finite MLE exists; an implementation must report that boundary rather than silently presenting a capped value as the MLE.

9 Complete algorithm

Inputs: matched training matrices (Y_1, Y_2) , survival (t, δ) , chosen K , and numerical tolerances. Feature order and preprocessing are recorded for subsequent prediction.

1. **Initialize.** Choose finite, nondegenerate score/loading moments, positive precisions, and priors with positive slab mass and rate. Initialize β and its prior away from an absorbing zero state, then compute projection moments using Equation 10.
2. **Refresh the survival approximation.** Set $\eta^0 = \mu_\eta$, compute (u, w, h) from Equation 8 and Equation 9, and freeze these quantities for the inner updates.
3. **For each factor k , update in order:**
 - Form residuals excluding k using current moments (Equation 13).
 - Update $q(L_{\cdot k})$ and $g_{L_{\cdot k}}$ using Equation 15 and Equation 16.
 - For each modality m , update $q(f_{mjk})$ sequentially using Equation 20, refreshing affected projection moments after each coordinate; then update $g_{F_{mk}}$ using Equation 22.
 - Recompute the current other-factor predictor and update g_{β_k} and $q(\beta_k)$ using Equation 24 through Equation 26.
4. **Update noise.** Compute expected residuals and update all τ_{mj} using Equation 28 and Equation 29.
5. **Assess convergence.** At fixed (w, h) , sweep until the working objective and moments stabilize, then refresh the survival approximation. Stop when outer iterations stabilize the predictor and moments; report failures and record the fixed objective and mean-predictor Cox partial likelihood as a diagnostic, not $\mathbb{E}_q \ell_C$.

No per-iteration column normalization is part of these equations. Factor labels and nonzero scales are exchangeable: rescaling a score column, both loading columns, and its survival coefficient can leave reconstruction and the predictor unchanged; estimated prior scales transform accordingly. Reporting may standardize a fitted nonzero score column after convergence with the corresponding parameter transformations, but this is not a fitting update.

For independent validation, hold fitted parameters and preprocessing fixed and compute $\hat{\eta}_{\text{val}} = (Y_{1,\text{val}} \bar{F}_1 + Y_{2,\text{val}} \bar{F}_2) \bar{\beta}$ without refitting to validation outcomes.